03_Drag_Calculation (Linear-in-v Diagnostic): Approaches 4–6

Validation_Aero_Test

February 12, 2026

1 Purpose and status

This document reports the results of the *linear-in-v* model variants implemented in `Code/03_Drag_Calculation` `Approach5_PairedOpposites_Linear`, and `Approach6_PooledByConfig_Linear`.

These approaches are kept as a diagnostic/experimental extension. For the current dataset and interval selection, they can yield non-physical results (notably negative C_dA in multiple configurations), so they are **not recommended** for reporting the final aerodynamic coefficients.

2 Model

With positive deceleration defined as $a_{\text{dec}} = -\dot{v}$, the linear-in-$v$ model fits

$$a_{\text{dec}}(v) = k_2 v^2 + k_1 v + c.$$

The coefficients map to reported quantities via

$$C_dA = \frac{2mk_2}{\rho}, \qquad C_r = \frac{c}{g}.$$

3 Why the linear-in-v term is not sensible here

The extended model introduces an extra degree of freedom (k_1). In principle it can represent speed-proportional losses; in practice, for typical coastdown segments in this dataset it creates an *identifiability problem*:

- **Collinearity over short speed ranges:** within a limited v window, v and v^2 are strongly correlated, so the fit cannot robustly distinguish k_1 from k_2.
- **Bias absorption:** unmodeled effects (residual grade, wind, sensor bias, filtering artefacts) can be absorbed by the extra term rather than improving the physical estimate of k_2.
- **Non-physical k_2 (negative C_dA):** when the regression reallocates curvature to the linear term, the estimated k_2 can become negative, which implies negative aerodynamic drag area.

The presence of negative C_dA across several configurations in Approaches 4–6 is therefore treated as evidence that the linear-in-v extension is not reliably constrained by the current data and preprocessing choices.

4 Implemented approaches (with example plots)

Approaches 4–6 mirror Approaches 1–3, but use the extended model.

4.1 Approach 4: Per-run (linear-in-v)

`Code/03_Drag_Calculation/Approach4_PerRun_Linear/run.py`

- Fit $a_{\text{dec}} = k_2 v^2 + k_1 v + c$ per extracted interval.
- Convert k_2, c to $C_d A$ and C_r, then summarize per configuration.

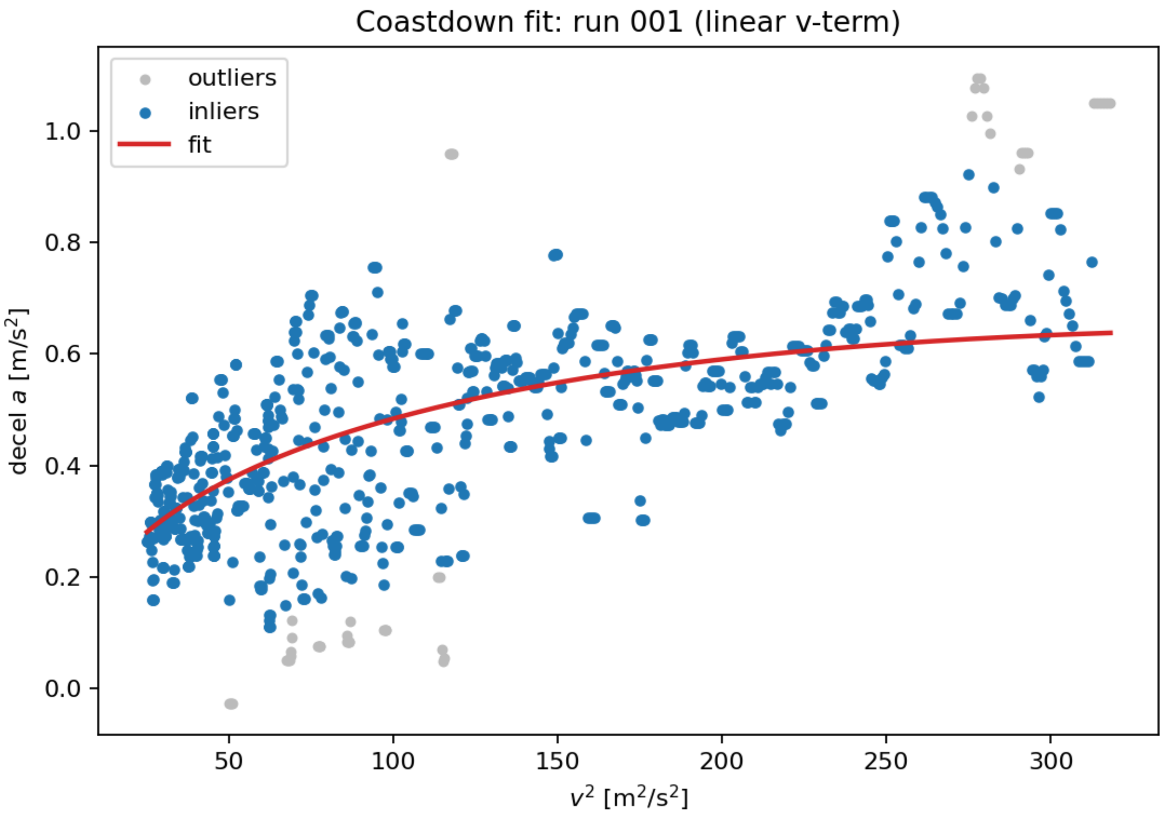

Figure 1: Approach 4: example per-run coastdown fit plot (Run 001) with linear-in-v term.

4.2 Approach 5: Paired opposites (linear-in-v)

`Code/03_Drag_Calculation/Approach5_PairedOpposites_Linear/run.py`

- Fit per run (as in Approach 4).
- Pair consecutive run IDs $(i, i{+}1)$ and average coefficients within each configuration.

4.3 Approach 6: Pooled by configuration (linear-in-v)

`Code/03_Drag_Calculation/Approach6_PooledByConfig_Linear/run.py`

- Stack all samples belonging to a configuration.
- Fit one (k_2, k_1, c) per configuration using robust regression.

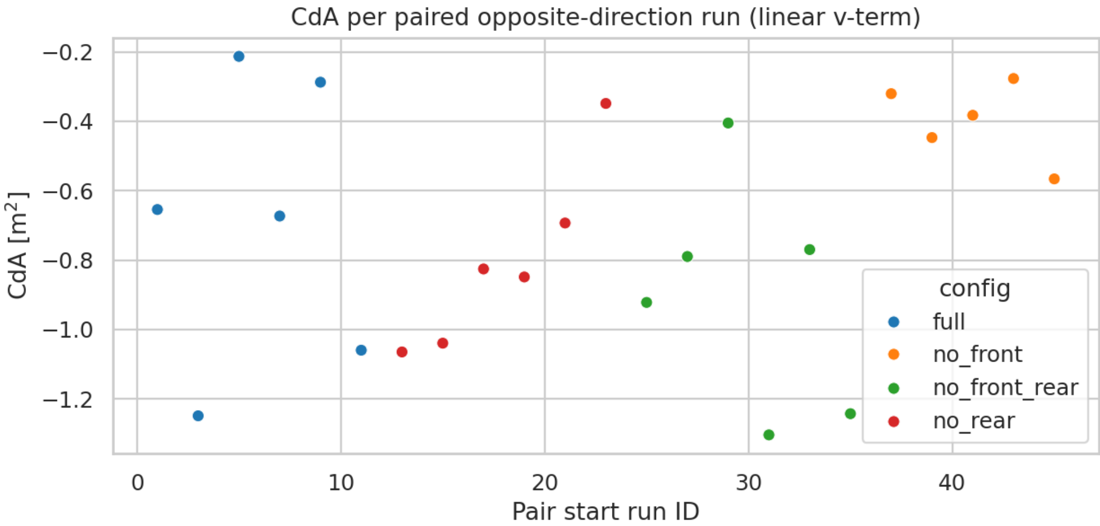

Figure 2: Approach 5: $C_d A$ per paired runs, grouped by configuration (linear-in-v variant).

5 Results (current Outputs/)

5.1 Uncertainty definition used in this report

The overview tables below report a "measurement uncertainty" per configuration:

- Approach 4: standard deviation across runs (`CdA_m2_std`, `Cr_std` from `drag_config_summary.csv`).
- Approach 5: standard deviation across paired runs.
- Approach 6: bootstrap standard deviation of the mean across runs (computed from Approach 4 per-run results and saved as `Outputs/03_Drag_Calculation/bootstrap_pooled_uncertainty.`

5.2 Overview: $C_d A$

Table 1: Linear-in-v model: $C_d A$ per configuration across Approaches 4–6 (mean/pooled value ± uncertainty).

Config	A4 (per-run)	A5 (paired)	A6 (pooled)
`full`	-0.690 ± 0.716	-0.690 ± 0.410	0.473 ± 0.200
`no_rear`	-0.804 ± 0.487	-0.804 ± 0.263	-0.024 ± 0.136
`no_front_rear`	-0.906 ± 0.569	-0.906 ± 0.333	-0.179 ± 0.161
`no_front`	-0.399 ± 0.176	-0.399 ± 0.114	0.252 ± 0.055

5.3 Overview: C_r

6 Reproducibility (example commands)

```
# Step 2 prerequisites:
# - drag intervals: Outputs/02_Run_Extraction/csv/drag_intervals.csv
# - per-run CSVs:   Outputs/02_Run_Extraction/csv/run_XXX.csv

python Code/03_Drag_Calculation/Approach4_PerRun_Linear/run.py
```

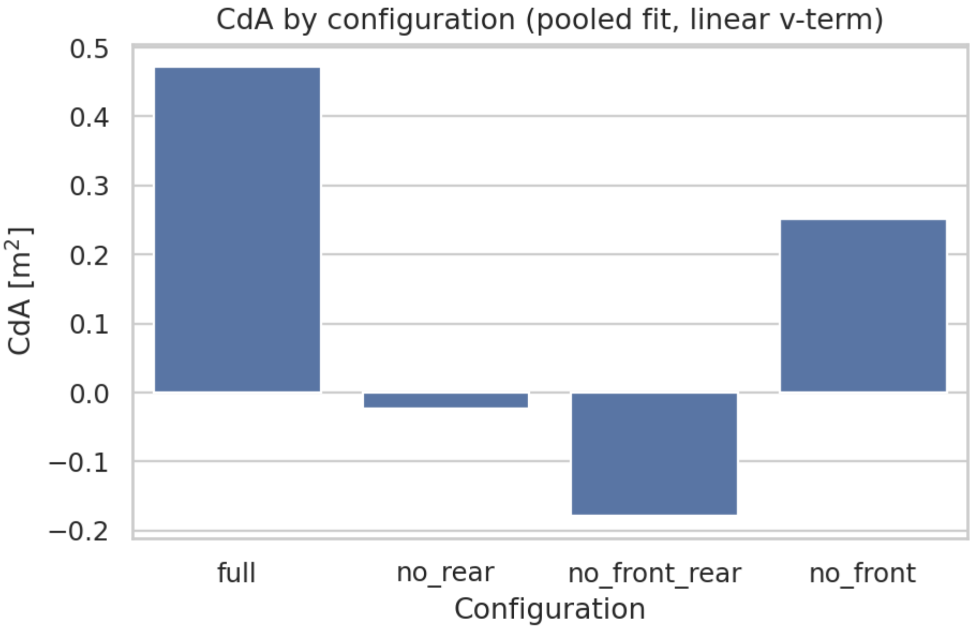

Figure 3: Approach 6: pooled C_dA per configuration (linear-in-v variant).

Table 2: Linear-in-v model: C_r per configuration across Approaches 4–6 (mean/pooled value $\pm$ uncertainty).

Config	A4 (per-run)	A5 (paired)	A6 (pooled)
`full`	-0.0153 ± 0.0277	-0.0153 ± 0.0134	0.0178 ± 0.0079
`no_rear`	-0.0166 ± 0.0177	-0.0166 ± 0.0100	0.0070 ± 0.0051
`no_front_rear`	-0.0133 ± 0.0153	-0.0133 ± 0.0079	0.0035 ± 0.0043
`no_front`	-0.0008 ± 0.0050	-0.0008 ± 0.0029	0.0113 ± 0.0015

```
python Code/03_Drag_Calculation/Approach5_PairedOpposites_Linear/run.py
python Code/03_Drag_Calculation/Approach6_PooledByConfig_Linear/run.py

# Optional: regenerate bootstrap uncertainty for pooled reporting
python Code/03_Drag_Calculation/bootstrap_pooled_uncertainty_from_runs.py
```